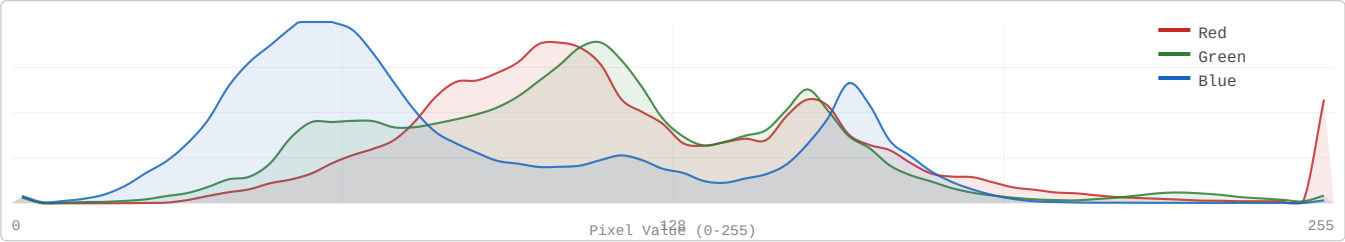
Decorrelation Gap: KODIM22

768x512 | 24-bit RGB | PNG (lossless)

Barn and Pond | Weakly one-dimensional (PC1 = 89.76%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.8714	0.8001
G	0.8714	1.0000	0.8752
B	0.8001	0.8752	1.0000

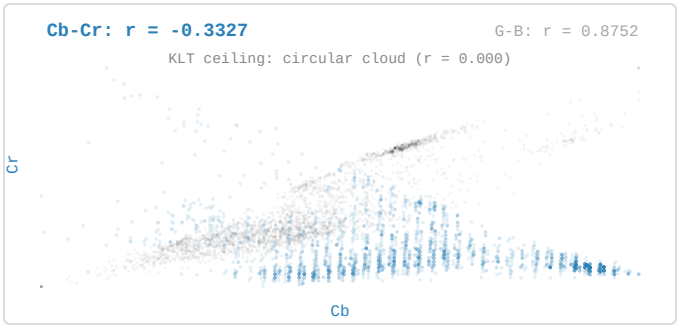
Cb-Cr: -0.333

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	0.0573	0.0595
Cb	0.0573	1.0000	-0.3327
Cr	0.0595	-0.3327	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.871394	0.057290	0.0573
R-B → Y-Cr	0.800057	0.059521	0.0595
G-B → Cb-Cr	0.875171	-0.332685	0.3327
Avg |r|	0.848874	0.149832	0.1498

Decorrelation Gap: KODIM22

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Barn and Pond | Weakly one-dimensional (PC1 = 89.76%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

82.3% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

10.5%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	42.55 dB
G	46.73 dB
B	40.50 dB
Overall	42.57 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5769	-0.5438	-0.6095	5232.62
PC2	-0.7151	-0.0243	0.6986	414.48
PC3	0.3947	-0.8389	0.3748	182.74

8. PCA Derived Metrics

Condition Number

28.63

PC1/PC2: 12.62 | PC2/PC3: 2.27

Blue Channel Independence

10.5%

Variance not captured by PC1

PC1 Dominant Channel

B

Loading: B (0.61)

9. Redundancy Profile

Classification: Weakly one-dimensional | CN: 28.63 | PC1: 89.8%

RGB Correlation: Avg |r| = 0.848874 | R-G: 0.8714 | R-B: 0.8001 | G-B: 0.8752

YCbCr Residual: Avg |r| = 0.149832 | Y-Cb: 0.0573 | Y-Cr: 0.0595 | Cb-Cr: -0.3327

Efficiency: 82.3% of KLT ceiling | Gap: 0.1498 avg |r| remaining

Blue Channel: 10.5% independent | 4:2:0 PSNR: 42.57 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.